

DATA 2010 Project Report: Trends in Olympic Athlete Performance and Representation (1896–2024)

Winter 2026

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1 Project Overview

Research question (what we are trying to find out). The Olympic Games provide a long-running record of athlete participation and medal outcomes, allowing us to study how performance and representation evolve over time. Using the *Olympic Summer and Winter Games (1896–2024)* dataset (athlete–event records with variables such as **year**, **season**, **city**, **sports**, **event**, **team**, **sex**, **age**, and **medal**), this project investigates three connected questions. First, we examine how medal outcomes vary by **age group** and **sex**, and how **gender parity** and **typical medal-winning age** changed across decades. Second, we evaluate how country participation size is associated with medal counts, medal efficiency, and Olympic dominance. Third, we examine which athlete-level factors most influence the probability of winning a medal and whether country/team membership adds predictive value after accounting for athlete characteristics.

Why it matters (who would care and why). The answers matter to stakeholders who make decisions about athlete development and sport investment. National sport organizations and coaches can use age- and sex-specific medal patterns to support talent pipeline planning. Analysts, journalists, and policymakers can benefit from fairer country comparisons by understanding how medal totals depend on delegation scale and efficiency rather than raw medal counts alone. Finally, understanding whether country-level advantages remain important after accounting for athlete characteristics can help explain why some countries outperform others over time.

Main result. Overall, the project shows that Olympic outcomes are shaped by both athlete-level and country-level factors. First, female participation increases substantially across decades and moves much closer to parity, while medalist composition and typical medal-winning ages vary across

age groups and sex. Second, country-year participation size is strongly positively associated with event-medal counts, but simple regression is not sufficient on its own; log transformation and multiple regression show that broader sport participation and time trends also help explain medal outcomes. Third, medal success is naturally a binary classification problem, and logistic regression shows that age, sex, sport, and country/team membership all contribute to an athlete-event row’s probability of winning a medal.

2 Research Question (Make it Specific)

The Olympic Games are not only a global competition but also a long-running dataset that can reveal how representation, performance, and strategic investment in sport evolve over time. Using Olympic records spanning 1896–2024, this project investigates patterns with real-world relevance to athlete development and national sport planning. Specifically, we focus on three research questions:

Our 3 Sub-questions

- 1) How do medal outcomes vary by age group and sex, and how did gender parity and typical medal-winning ages change across decades?
- 2) To what extent is country participation size associated with event-medal counts, and how does this relationship shape medal efficiency and Olympic dominance?
- 3) What factors most influence an athlete’s probability of winning a medal, and to what extent does country/team membership contribute to medal probability after accounting for athlete characteristics?

3 Data Source and Description

3.1 Where the data came from

- **Dataset name:** Olympic Summer and Winter Games (1896–2024)
- **Link (Kaggle):** <https://www.kaggle.com/datasets/magicchris/olympic-summer-and-winter-games-1896-2024>
- **Dataset owner (as shown on Kaggle):** Cassini-Chris (Owner)
- **Authors:** Not specified on the Kaggle dataset page
- **Provenance:** Not specified on the Kaggle dataset page
- **Citations / DOI:** Not specified on the Kaggle dataset page
- **What each row represents:** One row represents one athlete’s record in one Olympic event for a specific Olympic Games, including the athlete’s sex/age (if present), country/team, the sport and event, the Games year/season/city, and the medal outcome (if any).
- **Key columns used (2–8 columns):**
 - year
 - season
 - team
 - sports
 - event
 - sex
 - age / age_imputed
 - medal
- **License:** Apache License 2.0
- **Expected update frequency:** Not specified on the Kaggle dataset page

3.2 Basic dataset facts

Number of rows	290,697
Number of columns	8
Time range (Year)	1896–2024
Seasons included	Summer & Winter
Unique years	38
Unique teams (team)	1151
Unique sports (sports)	92
Unique events (event)	1534
Medalist rows	247,941
(Gold/Silver/Bronze)	
Age range (years)	10–72
Sex proportion (M/F)	69.38% / 30.62%
Missing values present?	Yes, mainly in age before cleaning
Exact duplicate rows removed	0

4 Data Cleaning and Feature Engineering

This section documents how we prepared the Olympics dataset for analysis and created new variables to support the three research questions.

Minimum expectations (what applies here):

- Fixed data types (e.g., **year** and **age** converted to numeric).
- Handled missing values (imputed missing **age** using group medians).
- Removed duplicates / impossible values (removed exact duplicate rows; dropped rows with invalid **year**).
- Dealt with outliers *carefully* (flagged suspicious ages instead of deleting them).

What we actually did:

- Standardized column names and stripped whitespace from text fields such as **team**, **sports**, **event**, and **medal**.
- Converted **year** and **age** to numeric form and removed rows with unusable year values.
- Standardized medal labels and created a binary variable **won_medal** indicating whether an athlete-event row resulted in Gold, Silver, or Bronze.
- Engineered **decade** to support long-run trend analysis.
- Engineered ordered **age_group** categories so age-based comparisons would be easier to interpret in plots and tables.

- Created `medal_points` for descriptive summaries comparing overall medal success.
- Cleaned detailed sport labels and grouped them into broader `sport_category` values to reduce inconsistency and improve interpretability in RQ3.
- Cleaned country/team labels and created `team_clean` so country-level aggregation would be interpretable.
- Created an *event-medal* view by deduplicating medals at (`year`, `event`, `team_clean`, `medal`) so that team events would not inflate country-level medal totals.
- Built country-year summaries including participation size, event-medal counts, efficiency, number of sports, and year for use in RQ2 regressions.

Show evidence (before vs. after):

Check	Before cleaning	After cleaning
<code>year</code> type	mixed / non-numeric possible	numeric integer
Missing <code>year</code>	present	removed
<code>age</code> type	mixed / invalid possible	numeric with imputation flag
Missing <code>age</code>	present	reduced by median imputation
Medal labels	inconsistent	standardized
Country/team labels	historical inconsistencies	cleaned <code>team_clean</code> to
Sport labels	overly detailed / inconsistent	standardized and grouped
Country medals	athlete-level inflation	event-medal view created

5 Exploratory Data Analysis and Visualization

Research Question 1: How do medal outcomes vary by age group and sex, and how did gender parity and typical medal-winning ages change across decades?

5.1 Plot 1: Medalist Share by Age Group (100% stacked by sex)

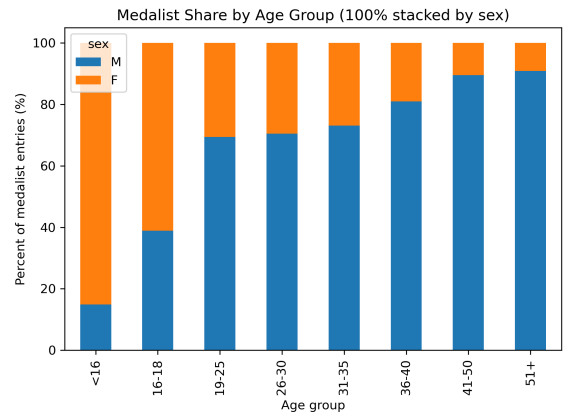


Figure 1: Each bar sums to 100%, showing the female vs. male share of medalist entries within each age group. This makes comparisons fair even when some age groups have far fewer medalist entries.

This figure indicates that sex balance among medalists varies by age group. Some age groups are more male-dominated, while others are closer to balanced. Differences are influenced by sport mix and historical changes in female participation.

Limitations: This uses medalist entries rather than all participants. Team events can inflate counts, and age imputation can slightly smooth patterns in smaller age bins.

5.2 Plot 2: Medal Type Share by Age Group (Gold/Silver/Bronze)

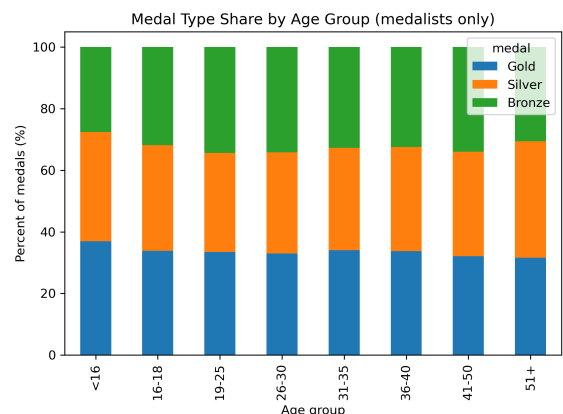


Figure 2: Each bar sums to 100%, showing how medals within each age group split into Gold, Silver, and Bronze.

This plot helps evaluate whether certain age groups have a relatively higher share of Gold compared to others. If medal-type shares are similar across

age groups, it suggests age affects the frequency of medals more than the composition of medal types.

Limitations: Percent shares hide total medal volume. Medal type is influenced by sport and competition depth, so this is descriptive rather than causal.

5.3 Plot 3: Gender Parity Over Time (Female and Male share by decade)

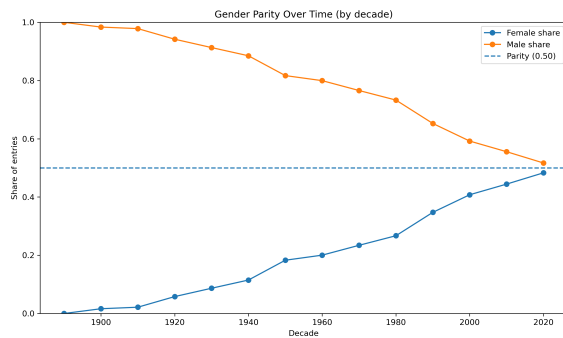


Figure 3: Female and male participation shares by decade. The dotted 0.50 line represents parity.

This plot shows a clear long-term increase in female participation over decades, moving toward parity. It provides strong evidence that Olympic representation has become more gender-balanced over time.

Limitations: This is participation share, not medal share. Dataset composition and coverage across early editions can affect decade-level estimates.

5.4 Plot 4: Largest Decade-to-Decade Changes in Female Share

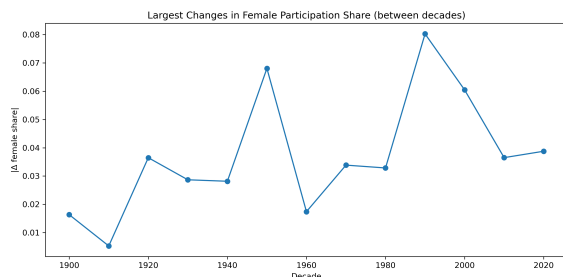


Figure 4: Absolute changes in female participation share between consecutive decades. Peaks identify decades with the largest shifts in gender representation.

This plot identifies the decades where gender representation changed the most. The highest peaks highlight key transition periods likely associated

with expanded women's events and broader inclusion policies.

Limitations: Decades with fewer Olympic editions or uneven coverage may exaggerate changes. Male share changes mirror female share by definition.

5.5 Plot 5: Overall Peak Medalist Age by Decade (Median + IQR, all sports)

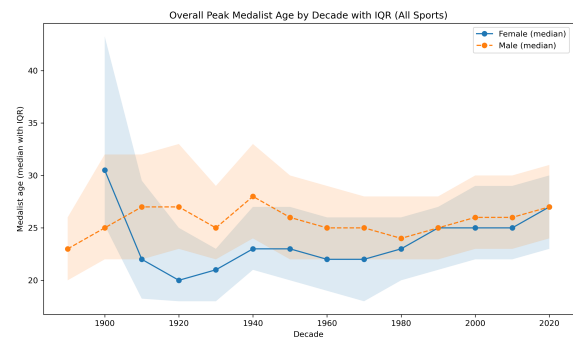


Figure 5: Median medalist age by decade with IQR shading (25th–75th percentile), computed across all sports combined and split by sex.

This figure summarizes how the typical medalist age and its spread change across decades for females and males. If medians rise, typical medal-winning ages trend older; if IQR widens, medalist ages become more varied.

Limitations: Combining all sports masks sport-specific peak ages. Team events can inflate medalist entries, and age imputation may slightly smooth the trend.

5.6 RQ1 Conclusion and Next Steps

Conclusion (RQ1): Overall, the results show that the Olympics become substantially more gender-balanced over time, with female participation increasing steadily and moving closer to parity in recent decades. Medalist composition varies across age groups and sex, and the peak-age plot indicates that typical medalist ages and their variability shift across decades.

Next steps (RQ1): With one more week, we would repeat the peak-age analysis by individual sports, separate Summer vs. Winter editions, include decade sample sizes on plots, and extend beyond EDA to a simple predictive model for medal probability.

Research Question 2: To what extent is country participation size associated with event-medal

counts, and how does this relationship shape medal efficiency and Olympic dominance?

5.7 Plot 6: Participation Size vs Event Medals with Least-Squares Fit

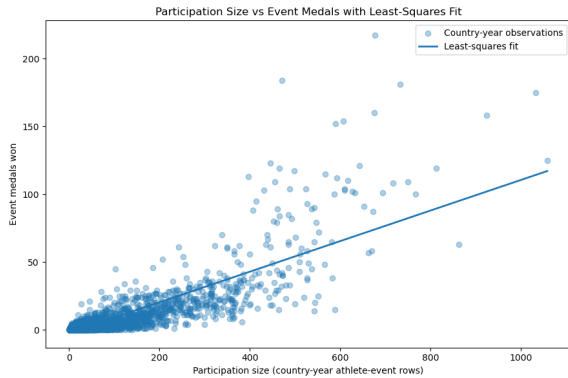


Figure 6: Scatterplot of country-year participation size versus event-medal counts, with a least-squares regression line summarizing the overall relationship.

This plot shows a strong positive association between participation size and event medals won. In general, country-years with larger delegations tend to win more medals.

The least-squares line captures this overall upward trend, but the points are fairly spread out around it. That spread becomes larger for bigger delegations, showing that countries with similar participation sizes can still have quite different medal totals. A few very large, high-medal observations also stand out, so the relationship is positive but not perfectly linear.

Limitations:

- Participation size is measured using athlete-event rows, not unique athletes.
- The variability increases at higher participation levels, so the fit is less consistent for large delegations.
- A small number of extreme country-year observations may pull the regression line upward.

5.8 Plot 7: Log-Transformed Participation Size vs Log-Transformed Event Medals

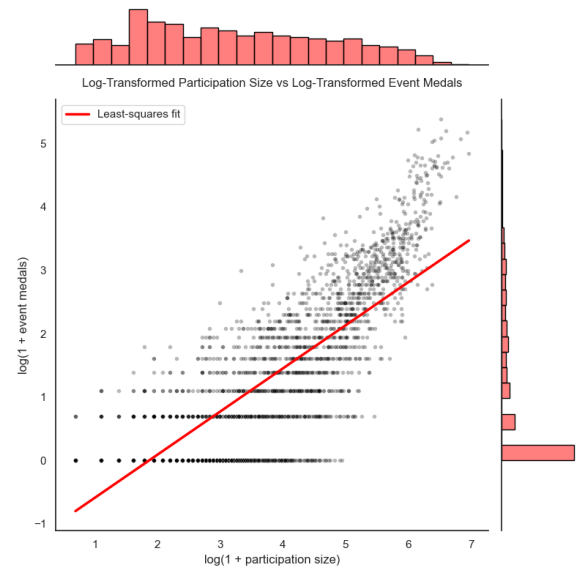


Figure 7: Log-transformed scatterplot showing the relationship between participation size and event-medal counts after reducing skewness and improving visibility at low values.

This log-transformed plot shows the relationship more clearly than the raw-scale scatterplot. After applying $\log(1 + \text{participation size})$ and $\log(1 + \text{event medals})$, the dense cluster of small country-year observations is spread out, making the overall pattern easier to see.

The points follow a clearer positive linear trend, and the least-squares line passes more centrally through the cloud. This suggests that the log transformation gives a better linear summary of the association between participation size and medal output. It also reduces the visual influence of extremely large delegations, so the relationship appears more stable across the full range of observations.

Overall, this figure supports the idea that participation size and medal success are positively associated, and that the relationship is represented more effectively on the log scale.

Limitations:

- There is still noticeable scatter around the fitted line, so participation size alone does not fully explain medal outcomes.
- The horizontal banding remains because medal counts are discrete values.
- The transformation improves linearity and visibility, but interpretation is now on the log

scale, not the original scale.

5.9 Plot 8: Residual Plot for the Multiple Regression Model

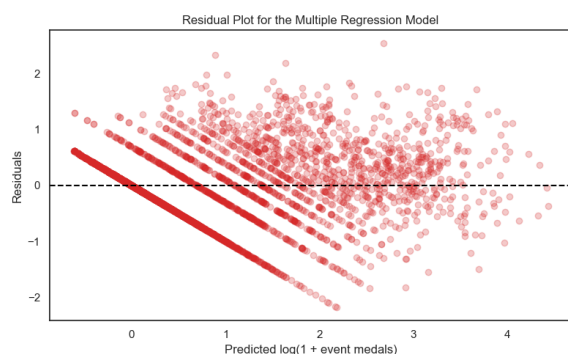


Figure 8: Residual plot from the multiple regression model used in RQ2(a), showing how prediction errors vary across fitted values.

This residual plot suggests that the multiple regression model improves on the simpler model, but it still does not fit perfectly. Many residuals are centered near zero, which is a good sign, and the spread is more balanced than in the one-predictor model.

At the same time, the residuals do not form a fully random cloud. There is clear diagonal banding and some remaining structure, especially across the middle range of predicted values, which shows that the model still leaves systematic variation unexplained. This supports the idea that adding predictors like participation size, number of sports, and year improves the fit, but does not completely capture medal outcomes.

Overall, the plot suggests that the multiple regression model is a better summary of the relationship for RQ2, though it is still only an approximation.

Limitations:

- The diagonal banding remains because the response is based on discrete medal counts, even after log transformation.
- Residual spread is still noticeable at moderate and higher fitted values, so the error variance is not fully constant.
- Important factors such as country strength, host advantage, and sport composition are not fully captured by the model.

5.10 Plot 9: Top Countries by Medal Efficiency

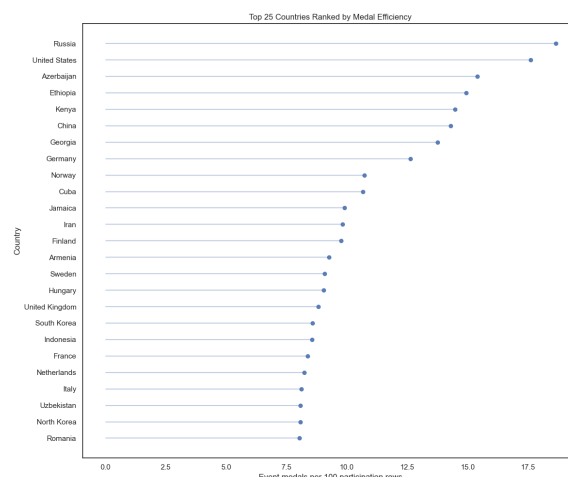


Figure 9: Countries ranked by medal efficiency, measured as event medals per 100 participation rows, after filtering out very small delegations.

This ranking compares countries by how efficiently they turn participation into medals after applying the minimum-size filter. Countries at the top win the most event medals per 100 participation rows, meaning they achieve stronger medal returns relative to the size of their Olympic participation.

The ranking shows that efficiency is not simply the same as delegation size. Some countries perform very well because they convert a moderate level of participation into a high medal yield, which may reflect strong specialization or consistently strong performances in the sports they enter. At the same time, highly efficient countries are not always the ones with the largest overall medal totals, so this plot highlights relative return rather than absolute dominance.

Overall, the figure suggests that some countries are especially effective at converting participation opportunities into medals.

Limitations:

- Efficiency is measured using athlete-event rows, not unique athletes.
- The ranking depends on the minimum participation threshold, so countries included or excluded can change the results.
- A high efficiency score does not necessarily mean broad success across many sports; it may reflect concentration in a smaller set of strong events.

5.11 Plot 10: Dominance in a Selected Olympic Year

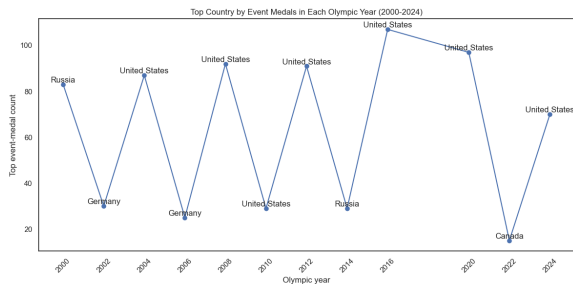


Figure 10: Country comparison for a selected Olympic year, showing how dominance relates to delegation size and medal efficiency.

This plot shows the country with the highest event-medal count in each Olympic year from 2000 to 2024. The pattern is clearly concentrated rather than widely shared: the United States appears most often as the top country, while Russia, Germany, and Canada lead in a smaller number of years.

The size of the top medal count also changes noticeably over time. Some years have much higher leading totals than others, which suggests that the strength of dominance is not constant across Olympic editions. Overall, the figure indicates that recent Olympic dominance has been driven by a small set of countries rather than rotating broadly across many nations.

Limitations:

- The dataset is missing the 2018 Winter Olympics, so the timeline is incomplete.
- The plot shows only the top country in each year, not how far ahead it was of the next countries.
- Summer and Winter Olympics are combined, so top medal counts are not directly comparable across all years.

5.12 RQ2 Conclusion and Next Steps

Conclusion (RQ2): Overall, the results show that Olympic medal success is strongly associated with participation size, with countries that enter more athlete-event rows generally winning more event medals. The log-transformed analysis makes this relationship clearer and more stable, while the multiple regression suggests that participation alone is not enough to explain medal outcomes fully. The efficiency rankings show that some countries convert

participation into medals especially well, but the year-by-year dominance plot suggests that repeated leadership is still concentrated among a small group of countries. Taken together, the results suggest that both scale of participation and efficiency of conversion matter, but large and historically strong programs remain the most likely to dominate medal tables over time.

Next steps (RQ2): With one more week, we would separate Summer and Winter Olympics, compare top-country dominance using relative gaps rather than just first-place counts, and test alternative models better suited for count outcomes. We would also add predictors such as host-country status, sport mix, and country fixed effects to better explain why countries with similar participation sizes can have very different medal returns.

Research Question 3: What factors most influence an athlete’s probability of winning a medal, and to what extent does country/team membership contribute to medal probability after accounting for athlete characteristics?

5.13 Plot 11: Medal Rate by Grouped Sport Category and Sex

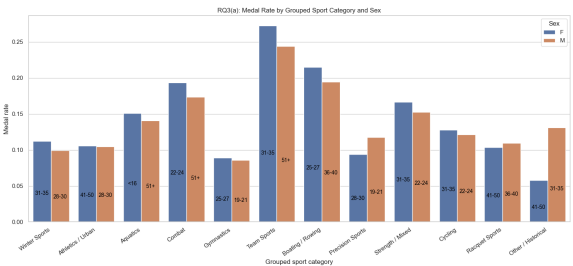


Figure 11: Medal rate by grouped sport category and sex, with dominant medal-winning age-bin labels added to each bar.

This plot shows that medal rates differ clearly across grouped sport categories and by sex. Team Sports, Boating/Rowing, and Combat have some of the highest medal rates, while categories such as Other/Historical, Precision Sports, and Gymnastics are lower. In most categories, the female and male bars are fairly close, but some gaps remain, showing that the relationship between sex and medal rate is not identical across all sports.

The age labels also suggest that the most common medal-winning age range varies by sport category. For example, some sports peak in younger adult ranges, while others are more concentrated in older age bins. This supports the idea that medal proba-

bility depends not just on sex, but also on the type of sport and the age profile typical for success in that sport.

Limitations:

- The analysis is based on athlete-event rows, not unique athletes.
- Grouping sports improves clarity, but it hides differences within each category.
- The plot is descriptive; the size of a bar difference alone does not show whether the difference is statistically meaningful.

5.14 Plot 12: ROC Curves for Logistic Regression Models

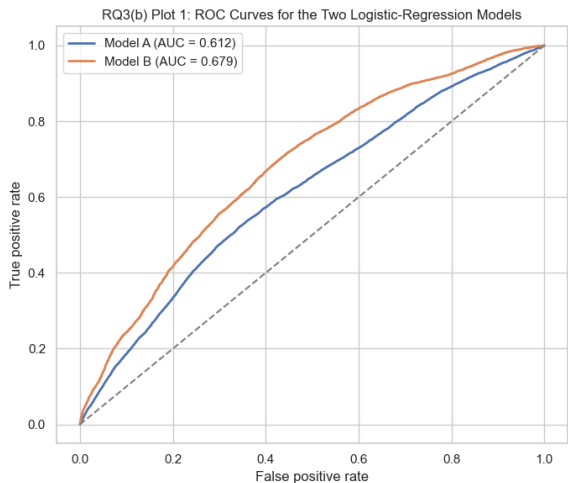


Figure 12: ROC curves comparing the athlete-level logistic regression model against the extended model that includes country/team membership.

The ROC curve for Model B stays above the curve for Model A, and its AUC is higher. This shows that once country/team is added to age, sex, and sport, the model does a better job distinguishing medal-winning from non-medal athlete-event rows.

The improvement is noticeable but not dramatic: both models perform better than random guessing, while Model B provides the stronger overall discrimination. This suggests that country/team membership contains meaningful information about medal probability beyond individual athlete characteristics alone, which directly supports the country-adjusted analysis later.

Limitations:

- ROC curves summarize overall classification

performance, but they do not show which specific countries drive the improvement.

- The AUC values are still moderate, so even the better model has limited predictive power.
- Better discrimination does not imply causation; it only shows improved separation between medal and non-medal outcomes.

5.15 Plot 13: Adjusted Country/Team Effects from Logistic Regression

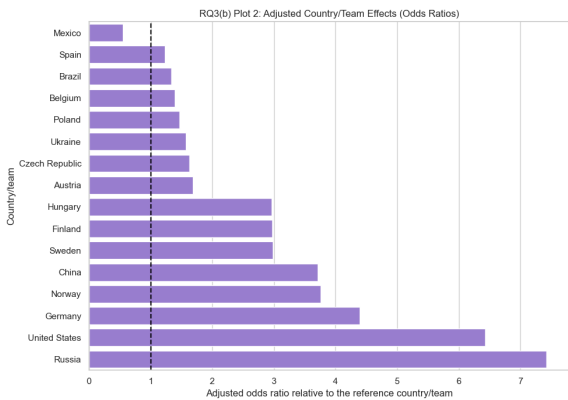


Figure 13: Selected country/team effects from the logistic regression model after controlling for age, sex, and sport category.

This plot shows the adjusted country/team effects from the logistic model. Countries with odds ratios above 1 have higher medal odds than the reference country/team even after controlling for age, sex, and sport category, while countries below 1 have lower adjusted medal odds.

The main result is that country differences still remain after athlete-level characteristics are taken into account. Some countries, such as Russia, the United States, and Germany, show substantially higher adjusted odds, while others fall closer to or below the reference line. This suggests that medal success depends not only on individual athlete characteristics, but also on country-level advantage that remains visible after adjustment.

Overall, the figure provides strong evidence that country/team membership adds important information about medal probability.

Limitations:

- The odds ratios are relative to a chosen reference country/team, so the values are comparative rather than absolute.

- These are model-based associations, not causal effects.
- The plot includes only a restricted set of more frequent countries/teams, so it does not represent every country in the dataset.

5.16 RQ3 Conclusion and Next Steps

Conclusion (RQ3): The results show that medal success is influenced by both athlete-level and country-level factors. Age, sex, and sport category all have clear associations with the probability of winning a medal, reflecting differences in peak performance ages and competitive structures across sports. The logistic regression results confirm that medal outcomes are appropriately modeled as a binary classification problem. Importantly, extending the model to include country/team membership improves classification performance, as seen in the higher ROC curve and AUC. This indicates that country-level factors—such as training systems, funding, and historical strength—continue to play a meaningful role in medal success even after accounting for individual athlete characteristics.

Next steps (RQ3): With additional time, we would extend the analysis in several ways. First, we would include interaction terms (e.g., age $\times$ sport category) to capture more detailed relationships between athlete characteristics and medal probability. Second, we would run separate models for major sport groups to account for differences in competitive structure and peak performance ages. Third, we would improve model evaluation by reporting additional metrics such as precision, recall, and optimal classification thresholds. Finally, we would incorporate more country-level variables, such as host-country effects and sport specialization, to better explain why countries with similar athlete profiles can have different medal outcomes.

5.17 Conclusion to EDA

Overall, the exploratory analysis shows that Olympic outcomes are shaped by long-term changes in representation, athlete demographics, and country-level participation patterns. Gender parity improves substantially across decades, medalist composition varies by age group and sex, and median medal-winning ages shift over time. At the country level, larger delegations generally win more event medals, but efficiency differs noticeably across countries, showing that participation size alone does not fully explain Olympic success.

These exploratory results motivate the use of regression and logistic-regression models in the next stage of the analysis.

6 Data Analysis (Techniques From This Course)

6.1 Analysis Plan

We first cleaned and standardized the Olympics dataset and engineered helper variables such as decade, age groups, binary medal indicators, cleaned country/team names, cleaned sport categories, and an event-medal view for country-level analysis. For RQ1, we used descriptive summaries and visualizations—especially 100% stacked bar charts, decade trend plots, and median/IQR age plots—to compare medal outcomes by age group and sex and to track gender parity and peak-age patterns over time. For RQ2, we aggregated the data to the country-year level and used scatterplots, least-squares regression, log transformation, residual plots, and then multiple regression with `log_participation_size`, `num_sports`, and `year` to model event-medal counts. For RQ3, we followed the course logistic-regression framework and treated medal success as a binary classification problem, modeling `won_medal` using athlete-level predictors such as age, sex, and sport category, and then extending the model with country/team membership. These methods are appropriate because they combine interpretable EDA with regression and classification models that directly match the three research questions.

6.2 Baseline (If you do prediction)

For RQ2, the baseline model is a simple least-squares regression using country-year participation size alone to predict event-medal counts. This gives a clear first summary of the positive association between delegation scale and medal success.

For RQ3, the baseline classifier is an athlete-level logistic regression model using age, sex, and sport category. We then compare this against an extended logistic regression model that also includes country/team membership to test whether country-level advantage improves prediction.

6.3 Main Results

The analysis supports all three research questions. For RQ1, the decade-level parity plots show a sub-

stantial increase in female participation over time, moving much closer to parity in recent decades. Medalist composition also varies by age group and sex, and the median/IQR age plots show that typical medal-winning ages shift across decades.

For **RQ2**, country-year participation size is strongly and positively associated with event-medal counts. The raw scatterplots and least-squares fit show the overall upward relationship, but the raw-scale pattern is highly skewed. Log transformation improves visibility and linearity, and the multiple regression model using `log_participation_size`, `num_sports`, and `year` fits better than the one-predictor model. The actual-versus-predicted and residual plots show that the multiple regression explains more of the medal pattern, although some structure and unexplained variation remain. Efficiency and dominance plots also show that countries with similar delegation sizes can differ meaningfully in their ability to convert participation into medals.

For **RQ3**, logistic regression is the appropriate course-aligned method because medal success is binary. The athlete-level model shows that age, sex, and sport category are associated with medal probability, and the extended model with country/team membership improves classification performance further. This indicates that athlete characteristics matter, but country-level advantage still contributes to medal probability after those characteristics are taken into account.

Model / Method	Main metric	Main takeaway
Simple regression	RMSE / R^2	Positive but limited first summary
Multiple regression	Improved R^2	Better fit than simple regression
Logistic regression	AUC / Accuracy	Athlete traits and country improve prediction

If you used a train/test split: We used a train/test split for the logistic-regression classification stage in RQ3 so that model performance was assessed on held-out data rather than training data only.

6.4 Interpretation (So What?)

In plain language, the results show that Olympic success is shaped by both representation and structure. At the athlete level, medal probability depends on age, sex, and sport category. At the country level, nations that send larger delegations tend to win more event medals, but countries with similar participation sizes can still differ substantially in efficiency and overall performance.

One important finding is that simple models are useful for showing the broad trend, but they are not sufficient on their own. In RQ2, the residual structure shows that participation size alone does not fully explain medal counts, which justifies moving to a multiple regression model with additional predictors such as number of sports and year. In RQ3, logistic regression is more appropriate than linear regression because medal success is a binary outcome.

The main limitations are that the dataset does not include a unique athlete identifier, so participation size is measured using athlete-event rows rather than exact unique athletes, and team events can inflate medal totals unless the event-medal method is used. In addition, combining many sports together can mask sport-specific differences in age patterns and medal probability.

7 Real-World Application

Stakeholders. The main stakeholders are national Olympic committees, sport federations, coaches, athlete development programs, and sports analysts.

What decision could they make based on your results? They could use these findings to make better decisions about talent development, event specialization, and how to evaluate country success more fairly than by medal totals alone.

Recommendations.

- **Use age- and sex-specific development pipelines:** Since medal outcomes and peak ages vary by age group and sex, athlete development should be tailored rather than based on a single generic model of peak performance.
- **Do not rely only on medal totals:** Because participation size strongly affects event-medal counts, countries should also be evaluated using medal efficiency and broader participation context.
- **Combine athlete-level and country-level planning:** Logistic regression results suggest that country/team membership still adds predictive value after accounting for athlete characteristics, so structural support systems matter alongside individual talent.

Confidence. We are moderately confident in these recommendations because the major relationships

are consistent across multiple plots and models, although the results remain observational and depend on the structure of the dataset.

8 Report Quality and Reproducibility

8.1 How to run the code

- The project code is available on GitHub: <https://github.com/PrinceKanani16/Data2010-olympics-project>
- Main file/notebook: `notebooks/Data2010.Project.ipynb`
- Steps to run:
 - Clone or download the GitHub repository.
 - Open the main Jupyter notebook.
 - Install the required Python packages if needed.
 - Run the notebook cells in order, starting from data loading and cleaning through feature engineering, visualization, and analysis.
 - Ensure the figure output folders exist or are created by the notebook before exporting plots.
- Required packages: `pandas`, `numpy`, `matplotlib`, `seaborn`, `scikit-learn`, `statsmodels`

8.2 Project Structure

- `README.md` (project overview and instructions for running the code)
- `notebooks/` (main analysis notebook)
- `data/` (dataset used in the project)
- `figures/` (exported plots organized by research question)
- `outputs/` (tables and model outputs)
- `report/` (final written report)

The project was organized into separate folders for the dataset, analysis notebook, exported figures, generated outputs, and final report to improve readability and reproducibility. The main analysis was completed in a Jupyter notebook, while figures were saved into subfolders by research question. Tables and model outputs were stored separately so they could be referenced more easily in the report and appendix.

```
DATA2010-Olympics-Project/
|
|-- README.md
|-- notebooks/
|   \-- Data2010_Project.ipynb
|
|-- data/
|   \-- olympics_1896_2024.csv
|
|-- figures/
|   |-- RQ1/
|   |-- RQ2/
|   |-- RQ3/
|   \-- Appendix/
|
|-- outputs/
|   |-- tables/
|   \-- model_outputs/
|
\-- report/
    \-- DATA2010_Project_Report.tex
```

9 Contributions (If Group Project)

- Prince Kanani: data cleaning, feature engineering, RQ1 structure, regression and logistic-regression setup, interpretation writing, final report integration.
- Joshua Mudiwa: plot integration, report formatting, figure organization, review of written sections, support on report structure and presentation.

10 Conclusion and Next Steps

Conclusion: This project shows that Olympic performance and representation changed substantially from 1896 to 2024. Female participation moved much closer to parity over time, medalist composition varied across age and sex groups, and typical medal-winning ages shifted across decades. At the country level, larger delegations generally won more event medals, but efficiency and dominance varied considerably, showing that participation size alone does not fully explain medal success. Finally, athlete-level logistic regression showed that medal probability depends on age, sex, and sport, and that country/team membership still improves prediction, indicating that country-level advantage remains important after athlete characteristics are taken into account.

Next steps: If we had one more week, we would improve the project in five ways:

- separate Summer and Winter Games more systematically
- extend the peak-age and medal-probability analyses within individual sport groups,
- report a fuller set of logistic-regression performance metrics and threshold choices,
- test additional country-level predictors such as host-country effects or more refined sport-specialization variables.
- compare team sports and individual sports more explicitly, since medal structures differ a lot between them

11 AI Disclosure

- We primarily used ChatGPT for generating prompts, checking code, and supporting analysis, with limited use of Claude for additional code review and analysis.
- We used them for:
 - Olympic dataset cleaning, organization, and presentation
 - Choosing the most appropriate plots for each research question to improve visualization
 - Improving graph readability through better labels, captions, spacing, and overall presentation
 - Clarifying sub questions so they were logical, distinct, and flowed well with the report and analysis
 - Ensuring the correct application of class techniques and methods
 - Using prompts to strengthen the final report’s writing, structure, and formatting
- Some examples of the prompts we used were the following:
 - Which of these research questions would be better for our project?
 - Where is the best place in this project to use a violin plot?
 - Improve this sentence for the report.
 - Fix spacing and formatting issues throughout the document.

References

- Olympic Summer and Winter Games (1896–2024). Kaggle dataset. Available at: <https://www.kaggle.com/datasets/magicchris/olympic-summer-and-winter-games-1896-2024>. Accessed March 25, 2026.
- Course notes: DATA 2010 Pandas Lecture (pandas1 Jupyter notebook and Pandas I lecture notes), University of Manitoba, Winter 2026.
- Course notes: DATA 2010 Lecture 14 (Pandas II), University of Manitoba, Winter 2026.
- Course notes: DATA 2010 Lecture 16 (Logistic Regression), University of Manitoba, Winter 2026.
- Course notes: DATA 2010 Lab 4 (Logistic Regression), University of Manitoba, Winter 2026.
- pandas documentation. Available at: <https://pandas.pydata.org/>
- scikit-learn documentation. Available at: <http://scikit-learn.org/stable/>
- statsmodels documentation. Available at: <https://www.statsmodels.org/>

A Appendix

Extra figures, extra tables, or small code snippets that would clutter the main report.

A.1 Top 15 Countries after Cleaning the Dataset

Table 1: Top countries by participant count after dataset cleaning

Country	Count
United States	16681
Germany	14396
France	10756
United Kingdom	10691
Russia	10042
Italy	9722
Canada	9057
Japan	8212
Australia	7566
Sweden	7368
Hungary	6042
Poland	5890
Czech Republic	5549
Switzerland	5534
Netherlands	5440

A.2 Total Number of Participants in Each Olympic Sport Category

Table 2: This table shows the total number of participants in each Olympic sport category after cleaning the dataset and regrouping individual sports into broader categories (Feature Engineering). It highlights how athlete participation is distributed across categories, with Winter Sports and Athletics / Urban having the largest counts.

Sport Category	Count
Winter Sports	40548
Athletics / Urban	39563
Aquatics	29142
Combat	27964
Gymnastics	22240
Team Sports	21994
Boating / Rowing	20248
Precision Sports	13632
Strength / Mixed	11905
Cycling	10338
Racquet Sports	6725
Other / Historical	1545

A.3 Proportion of Olympic Medals Won by Continent

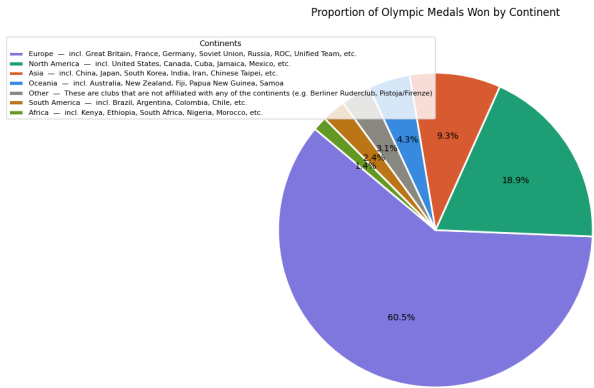


Figure 14: Supplementary figure showing the distribution of medal-winning entries across continents. This figure is included in the appendix because it provides global context but is not central to the main three research questions.